
Department of Statistics & Computer Science
University of Kelaniya
Academic Year -2024/2025 (Semester II)
COSC 32133 Full-Stack Software Development
Lab Sheet 01
Student No – PS/2022/033

HTML code

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>SkyCast - Weather Dashboard</title>
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <div class="page">

<header class="site-header">

  <div class="brand">

    <div class="logo"><img alt="Weather icon" data-bbox="325 555 345 575"/></div>

    <div>
      <h1>SkyCast</h1>
      <p>Weather made simple</p>
    </div>

  </div>

  <div class="header-badge"><img alt="Globe icon" data-bbox="385 715 405 735"/> Live Weather</div>
</header>

<main>

  <section class="hero">

    <h2>Check the weather anywhere</h2>
```

```

<p class="subtitle">
  Search any city for current weather and a 5-day forecast.
</p>

<div class="search-row">

  <input
    type="text"
    id="cityInput"
    placeholder="Enter a city, e.g. Colombo"
  >

  <button id="searchBtn">Search</button>

</div>

<p id="statusMsg" class="status"></p>
</section>

<section id="currentCard" class="current-card hidden">

  <h2 id="cityName">—</h2>

  <div class="current-grid">

    <div class="stat">
      <span class="stat-label">Temperature</span>

      <span class="stat-value" id="temp">—</span>
    </div>

    <div class="stat">
      <span class="stat-label">Wind Speed</span>

      <span class="stat-value" id="wind">—</span>
    </div>

    <div class="stat">
      <span class="stat-label">Condition</span>

      <span class="stat-value" id="condition">—</span>
    </div>
  </div>

```

```

    </div>

</div>

</section>

<section class="forecast-section">

  <div class="section-heading">

    <div>

      <h2>5-Day Forecast</h2>

      <p>Upcoming weather conditions</p>

    </div>

    <span class="forecast-icon">📅</span>

  </div>

  <div class="table-wrapper">

    <table>

      <thead>
        <tr>
          <th>Date</th>
          <th>Condition</th>
          <th>Max °C</th>
          <th>Min °C</th>
          <th>Precipitation (mm)</th>
        </tr>
      </thead>

      <tbody id="forecastBody">
      </tbody>

    </table>

  </div>

```

```
</section>

</main>

<footer class="site-footer">

  <div class="footer-brand">
     <strong>SkyCast</strong>
  </div>

  <p>Simple, fast and reliable weather information.</p>

  <div class="footer-bottom">

    <span>© 2026 SkyCast</span>
    <span>Powered by Open-Meteo</span>

  </div>

</footer>

</div>
<script src="script.js"></script>
</body>
</html>
```

CSS code

```
:root {
  --accent: #0e7c86;
  --accent-dark: #095b62;
  --accent-light: #e5f5f6;

  --bg: #f4f7f8;
  --card: #ffffff;
  --border: #dde3e5;

  --text: #1c2130;
  --muted: #5b6273;

  --rain: #e3f2fd;
}

* {
  box-sizing: border-box;
}

body {
  margin: 0;
  font-family: "Segoe UI", -apple-system, BlinkMacSystemFont,
    Roboto, Helvetica, Arial, sans-serif;

  background:
    radial-gradient(circle at top right, #e3f5f6 0, transparent 35%),
    var(--bg);

  color: var(--text);
}

.page {
  max-width: 760px;
  margin: 0 auto;
  padding: 0 20px 40px;
}

.site-header {
  display: flex;
  justify-content: space-between;
  align-items: center;

  padding: 22px 0;
```

```

margin-bottom: 35px;

border-bottom: 1px solid var(--border);
}

.brand {
  display: flex;
  align-items: center;
  gap: 12px;
}

.logo {
  width: 48px;
  height: 48px;

  display: flex;
  align-items: center;
  justify-content: center;

  background: var(--accent-light);
  border-radius: 14px;

  font-size: 1.6rem;
}

.brand h1 {
  margin: 0;
  font-size: 1.45rem;
  letter-spacing: -0.02em;
}

.brand p {
  margin: 2px 0 0;
  color: var(--muted);
  font-size: 0.8rem;
}

.header-badge {
  padding: 8px 13px;

  background: var(--accent-light);
  color: var(--accent-dark);

  border-radius: 20px;

```

```
    font-size: 0.78rem;
    font-weight: 600;
  }

  .hero {
    text-align: center;
    margin-bottom: 25px;
  }

  .hero h2 {
    margin: 0 0 8px;

    font-size: 2rem;
    letter-spacing: -0.03em;
  }

  .subtitle {
    color: var(--muted);
    margin: 0 0 25px;
  }

  .search-row {
    display: flex;
    gap: 10px;

    max-width: 600px;
    margin: 0 auto 10px;
  }

  #cityInput {
    flex: 1;

    padding: 14px 16px;

    border: 1px solid var(--border);
    border-radius: 10px;

    font-size: 1rem;

    outline: none;

    background: white;

    transition: border 0.2s ease, box-shadow 0.2s ease;
  }
```

```

#cityInput:focus {
  border-color: var(--accent);

  box-shadow: 0 0 0 3px rgba(14, 124, 134, 0.12);
}

#searchBtn {
  padding: 14px 24px;

  border: none;
  border-radius: 10px;

  background: var(--accent);
  color: #fff;

  font-size: 1rem;
  font-weight: 600;

  cursor: pointer;

  transition:
    background 0.2s ease,
    transform 0.15s ease,
    box-shadow 0.2s ease;
}

#searchBtn:hover {
  background: var(--accent-dark);

  transform: translateY(-1px);

  box-shadow: 0 5px 14px rgba(9, 91, 98, 0.2);
}

#searchBtn:active {
  transform: translateY(0);
}

.status {
  min-height: 20px;

  color: var(--muted);

  font-size: 0.92rem;
}

```

```

    margin: 8px 2px 20px;
}

.status.error {
    color: #c0392b;
}

.current-card {
    background: var(--card);

    border: 1px solid var(--border);
    border-radius: 16px;

    padding: 25px;

    margin-bottom: 35px;

    box-shadow:
        0 8px 25px rgba(28, 33, 48, 0.06);

    animation: fadeIn 0.35s ease;
}

.current-card.hidden {
    display: none;
}

.current-card h2 {
    margin: 0 0 18px;

    font-size: 1.35rem;
}

.current-grid {
    display: grid;

    grid-template-columns: repeat(3, 1fr);

    gap: 14px;
}

.stat {
    display: flex;

```

```

flex-direction: column;

background: var(--bg);

border-radius: 12px;

padding: 16px;

text-align: center;

border: 1px solid transparent;

transition:
  transform 0.2s ease,
  border 0.2s ease;
}

.stat:hover {
  transform: translateY(-2px);

  border-color: var(--border);
}

.stat-label {
  font-size: 0.72rem;

  text-transform: uppercase;

  letter-spacing: 0.05em;

  color: var(--muted);

  margin-bottom: 6px;
}

.stat-value {
  font-size: 1.2rem;

  font-weight: 700;

  color: var(--accent-dark);
}

.forecast-section {
  margin-bottom: 40px;
}

```

```

}

.section-heading {
  display: flex;

  justify-content: space-between;
  align-items: center;

  margin-bottom: 14px;
}

.forecast-section h2 {
  font-size: 1.25rem;

  margin: 0 0 3px;
}

.section-heading p {
  margin: 0;

  color: var(--muted);

  font-size: 0.85rem;
}

.forecast-icon {
  width: 40px;
  height: 40px;

  display: flex;
  align-items: center;
  justify-content: center;

  background: var(--accent-light);

  border-radius: 10px;

  font-size: 1.15rem;
}

.table-wrapper {
  overflow-x: auto;

  border-radius: 13px;
}

```

```

    box-shadow:
      0 5px 20px rgba(28, 33, 48, 0.04);
  }

table {
  width: 100%;

  border-collapse: collapse;

  background: var(--card);

  border-radius: 13px;

  overflow: hidden;

  border: 1px solid var(--border);
}

th,
td {
  text-align: left;

  padding: 12px 13px;

  border-bottom: 1px solid var(--border);

  font-size: 0.88rem;
}

tbody tr:last-child td {
  border-bottom: none;
}

th {
  background: #eef5f5;

  text-transform: uppercase;

  font-size: 0.72rem;

  letter-spacing: 0.04em;

  color: var(--accent-dark);

  white-space: nowrap;

```

```

}

td {
  white-space: nowrap;
}

tbody tr {
  transition: background 0.15s ease;
}

tbody tr:hover {
  background: #f8fafb;
}

tr.rainy {
  background: var(--rain);
}

tr.rainy:hover {
  background: #dceefb;
}

.site-footer {
  margin-top: 55px;

  padding: 25px 0 5px;

  border-top: 1px solid var(--border);

  text-align: center;
}

.footer-brand {
  display: flex;

  justify-content: center;
  align-items: center;

  gap: 7px;

  font-size: 1.05rem;
}

.footer-brand span {
  font-size: 1.25rem;
}

```

```

}

.site-footer > p {
  margin: 8px 0 18px;

  color: var(--muted);

  font-size: 0.82rem;
}

.footer-bottom {
  display: flex;

  justify-content: center;

  gap: 20px;

  color: var(--muted);

  font-size: 0.75rem;
}

@keyframes fadeIn {
  from {
    opacity: 0;
    transform: translateY(8px);
  }

  to {
    opacity: 1;
    transform: translateY(0);
  }
}

@media (max-width: 600px) {

  .page {
    padding: 0 15px 30px;
  }

  .site-header {
    margin-bottom: 30px;
  }

  .header-badge {

```

```

    display: none;
}

.hero h2 {
    font-size: 1.6rem;
}

.search-row {
    flex-direction: column;
}

#searchBtn {
    width: 100%;
}

.current-grid {
    grid-template-columns: 1fr;
}

.current-card {
    padding: 20px;
}

th,
td {
    font-size: 0.78rem;
    padding: 10px;
}

.footer-bottom {
    flex-direction: column;
    gap: 5px;
}
}

```

JAVA SCRIPT code

```
const cityInput = document.getElementById("cityInput");
const searchBtn = document.getElementById("searchBtn");
const statusMsg = document.getElementById("statusMsg");
const currentCard = document.getElementById("currentCard");
const cityNameEl = document.getElementById("cityName");
const tempEl = document.getElementById("temp");
const windEl = document.getElementById("wind");
const conditionEl = document.getElementById("condition");
const forecastBody = document.getElementById("forecastBody");

function describeWeatherCode(code) {
  if (code === 0) {
    return "Clear sky";
  } else if (code >= 1 && code <= 3) {
    return "Partly cloudy";
  } else if (code === 45 || code === 48) {
    return "Fog";
  } else if (code >= 51 && code <= 57) {
    return "Drizzle";
  } else if (code >= 61 && code <= 67) {
    return "Rain";
  } else if (code >= 71 && code <= 77) {
    return "Snow";
  } else if (code >= 80 && code <= 82) {
    return "Rain showers";
  } else if (code >= 95 && code <= 99) {
    return "Thunderstorm";
  }

  return "Unknown";
}

function setStatus(message, isError = false) {
  statusMsg.textContent = message;

  if (isError) {
    statusMsg.classList.add("error");
  } else {
    statusMsg.classList.remove("error");
  }
}
```

```

async function geocodeCity(city) {
  const url =
    `https://geocoding-api.open-
meteo.com/v1/search?name=${encodeURIComponent(city)}&count=1`;

  const res = await fetch(url);

  if (!res.ok) {
    throw new Error("Geocoding request failed");
  }

  const data = await res.json();

  if (!data.results || data.results.length === 0) {
    throw new Error("City not found – try another name.");
  }

  return data.results[0];
}

async function fetchForecast(lat, lon) {
  const url =
    `https://api.open-meteo.com/v1/forecast?latitude=${lat}&longitude=${lon}`
+
    `&current_weather=true` +
    `&daily=weathercode,temperature_2m_max,temperature_2m_min,precipitation_s
um` +
    `&timezone=auto`;

  const res = await fetch(url);

  if (!res.ok) {
    throw new Error("Forecast request failed");
  }

  return res.json();
}

function renderCurrentWeather(place, weatherData) {
  const current = weatherData.current_weather;

  cityNameEl.textContent = `${place.name}, ${place.country}`;
  tempEl.textContent = `${current.temperature} °C`;
  windEl.textContent = `${current.windspeed} km/h`;
  conditionEl.textContent = describeWeatherCode(current.weathercode);
}

```

```

    currentCard.classList.remove("hidden");
}

function renderForecastTable(daily) {
    forecastBody.innerHTML = "";

    for (let i = 0; i < daily.time.length; i++) {
        const row = document.createElement("tr");

        row.innerHTML = `
            <td>${daily.time[i]}</td>
            <td>${describeWeatherCode(daily.weathercode[i])}</td>
            <td>${daily.temperature_2m_max[i]} °C</td>
            <td>${daily.temperature_2m_min[i]} °C</td>
            <td>${daily.precipitation_sum[i]} mm</td>
        `;

        if (daily.precipitation_sum[i] > 0) {
            row.classList.add("rainy");
        }

        forecastBody.appendChild(row);
    }
}

async function handleSearch() {
    const city = cityInput.value.trim();

    if (city === "") {
        setStatus("Please type a city name.", true);
        return;
    }

    currentCard.classList.add("hidden");
    forecastBody.innerHTML = "";

    setStatus("Loading...");

    try {
        const place = await geocodeCity(city);

        const weatherData = await fetchForecast(
            place.latitude,

```

```

        place.longitude
    );

    renderCurrentWeather(place, weatherData);
    renderForecastTable(weatherData.daily);

    setStatus("");
  } catch (error) {
    setStatus(error.message, true);
  }
}

searchBtn.addEventListener("click", handleSearch);

cityInput.addEventListener("keydown", function (e) {
  if (e.key === "Enter") {
    handleSearch();
  }
});

```

